a

Energy $E(x)$

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Deep, isolated wells: exact readout

Shallow, merged wells: approximate readout

n_{crit} $(\sigma_n = 1)$

Number frequency f_n

0.00 0.05 0.10 0.15 0.20 0.25 0.30

Evidence axis x (the number line a trial's state moves on)

b

$p(n \mid x^*)$, y_0=2 — sharp
$H = 0.02$

$p(n \mid x^*)$, y_0=6 — moderate
$H = 1.15$

$p(n \mid x^*)$, y_0=12 — broad
$H = 2.45$

before clean-up

clean-up steps

after clean-up

Expected readout entropy $E[H]$

Convergence steps $E[T]$

0 50 100 150 200

c

basal latency t_0
attractor time $\alpha E[T]$
readout uncertainty $\beta_H E[H]$

fast, exact

slow, approximate

$\text{RT} = t_0 + \alpha E[T] + \beta_H E[H]$

Predicted RT (s)

1.5 2.0 2.5 3.0

Presented numerosity n (one clean-up episode per trial)